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ORIGINAL PAPER

New paleomagnetic data from Late Neoproterozoic sedimentary successions in Southern Urals, Russia: implications for the Late Neoproterozoic paleogeography of the Iapetan realm

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Abstract We present the results of paleomagnetic study of Ediacaran sedimentary successions from the Southern Urals. The analysis of the sedimentary rocks of the Krivaya Luka, Kurgashlya and Bakeevo Formations reveal stable mid-temperature and high-temperature remanence components. Mid-temperature components were acquired during Devonian (Bakeevo Formation) and Late Carboniferous–Early Permian remagnetization events. The high-temperature components in Kurgashlya and Bakeevo Formations are interpreted to be primary, because they are supported by a positive conglomerate test (Bakeevo Formation) and magnetostratigraphic pattern (Kurgashlya Formation). The high-temperature component in the Krivaya Luka Formation is interpreted to be a Late Ediacaran overprint. Our new paleomagnetic poles together with some previously published Ediacaran poles from Baltica and Laurentia are used herein to produce a series of paleogeographic reconstructions of the opening of the Iapetus Ocean.

Keywords Paleomagnetism · Paleogeography · Ediacaran · Baltica · Laurentia · Urals · Iapetus

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Introduction

Several global-scale important events occurred during the last 100 m.y. of the Neoproterozoic time, including one or more major global-scale glaciations (e.g., Kirschvink 1992; Hoffman and Schrag 2002; Chumakov 2011), the explosion of Ediacaran fauna (Knoll 1992; McCall 2006), the final break-up of the Rodinia supercontinent by opening of the Iapetus Ocean, Tornquist Sea and Paleo-Asian Ocean (e.g., Bingen et al. 1998; Cawood et al. 2001; Didenko et al. 1994; Cawood and Pisarevsky 2006; Pisarevsky et al. 2008), and the assembly of the Gondwana supercontinent by closing the Mozambique, Adamastor and Brasiliano oceans (Pimentel et al. 1999; Collins and Pisarevsky 2005). The variety of Late Neoproterozoic paleogeographic reconstructions (e.g., Dalziel 1997; Cordani et al. 2003; Hartz and Torsvik 2002; Murphy et al. 2004, 2013; Collins and Pisarevsky 2005; Cawood and Pisarevsky 2006; McCausland et al. 2006; Tohver et al. 2006) is caused, in part, by the scarcity of Neoproterozoic paleomagnetic data (Pisarevsky 2005) and particularly by the controversy over the North American paleomagnetic data (e.g., McCausland and Hodych 1998; Pisarevsky et al. 2000, 2001; Pisarevsky et al. 2008 and references therein). These controversies greatly affect reconstructions of the Late Neoproterozoic breakup of Baltica and Laurentia during the opening of the Iapetus Ocean.

Paleomagnetic data from Baltica for this time are also scarce. Only few well-dated and reliable paleopoles are

available: the 616 ± 3 Ma Egersund dykes (Norway) pole (Walderhaug et al. 2007), the combined 555–550 Ma Winter Coast (Russia) pole from clastic rocks (Russia; Popov et al. 2002, 2005; Iglesia Llanos et al. 2005) and the recently published ~550 Ma pole from the sedimentary Zigan Formation of the Southern Urals (Levashova et al. 2013). Importantly, these poles have their Laurentian coeval counterparts: the 615 ± 2 Ma Long Range dykes pole (Murthy et al. 1992; Kamo and Gower 1994) and the 550 ± 3 Ma Skinner Cove Formation pole (McCausland and Hodych 1998; Cawood et al. 2001). The comparison of these pairs of coeval poles together with geological evidence (e.g., Compston et al. 1995; Nikishin et al. 1996; Bingen et al. 1998; Greiling et al. 1999; Siedlecka et al. 2004; Nosova et al. 2005) suggests that at ca. 615 Ma Baltica and Laurentia were still in their Rodinian fit (Pisarevsky et al. 2003, 2008; Li et al. 2008), but at ca. 550 Ma they were widely separated by the Iapetus Ocean (Cawood and Pisarevsky 2006). The details of the opening of Iapetus are still unknown. Some of Laurentian ca. 600–560 Ma poles (Van Alstine and Gillett 1979; Tanczyk et al. 1987; Symons and Chiasson 1991; Meert et al. 1994; McCausland et al. 2011) and some non-key Baltican poles (Meert et al. 1998; Nawrocki et al. 2004; Iosifidi et al. 2005; Elming et al. 2007) may support various scenarios. Most of these data are in favor of low-latitude position of Baltica between ca. 600 and 550 Ma. However, the details of its movement with respect to Laurentia are uncertain.

Here, we present new paleomagnetic data from Late Neoproterozoic sedimentary rocks in the western part of the Southern Urals. Some previous paleopoles from this area (e.g., Komissarova in Khramov 1971) are suspicious because of their closeness to the mid-Paleozoic part of the European apparent polar wander path (Smethurst et al. 1998), which may be caused by the well-known remagnetization event (e.g., Shipunov et al. 2007; Lubnina 2009). Levashova et al. (2013) recently reported a new high-quality paleomagnetic result from the Zigan Formation in South Urals—the uppermost part of the Ediacaran (Vendian) sedimentary succession in this area. We have concentrated on some older strata from the same area.

Geology and sampling

The Bashkirian mega-anticlinorium (BMA) is a major structure on the western slope of the Southern Urals in the southeastern corner of the Proterozoic continent, Baltica (Figs. 1, 2). The core of this structure is composed of thick siliciclastic and carbonate Mesoproterozoic (Lower and Middle Riphean), Neoproterozoic and Paleozoic sedimentary successions with some presence of mafic and felsic lavas. In some places, these rocks are intruded by gabbro-dolerites and granites (Puchkov 2010; Puchkov et al. 2013). There is a broad agreement that most of these sedimentary successions accumulated on a long-lived passive margin of Baltica at least in the Late Riphean (Cryogenian) with change to active margin conditions after the Early Vendian (Ediacaran) (e.g., Nikishin et al. 1996; Willner et al. 2001, 2003; Maslov and Isherskaya 2002; Puchkov 2003 and references therein; Bogdanova et al. 2008).

Vendian (Ediacaran) successions in BMA are represented by the Asha Group (Fig. 2). This group overlies the Upper Riphean (Cryogenian) sedimentary succession with erosional unconformity (Uk Formation in the northern part of BMA and Krivaya Luka Formation in the south, Fig. 2) and is unconformably overlain by Lower Devonian to Middle Ordovician sediments (Puchkov 2010). The Asha Group is subdivided into (from top to bottom) Zigan, Kuk-Karauk, Basu, Uryuk and Bakeevo Formations in the northern BMA and into Bainazarovo and Kurgashlya Formations in the southern BMA (Fig. 2). The Kurgashlya Formation is correlated to the Bakeevo Formation and both of them probably were accumulated in the passive margin environments (Willner et al. 2001). The metamorphic grade varies from low diagenetic to anchizonal (Matenaar et al. 1999). Both formations overlie Riphean strata with erosional unconformity and contain diamictite in their lower parts. Bainazarovo and Uryuk Formations are also considered as correlatives, they are both overlain by the Basu Formation of the foreland basin (Willner et al. 2001, 2003).

Bekker (1992) and Puchkov (2012a) reported the presence of Ediacaran fauna in the Zigan and Basu formations, which indicates a maximum age of ca. 580 Ma for these formations. This interpretation is supported by the 548 ± 4 Ma U–Pb age of the thin bed of volcanic tuffs in the lower Zigan Formation (Grazhdankin et al. 2011; Levashova et al. in press). However, Kuznetsov and Shazillo (2011) challenge this interpretation by reporting of small fragments of skeletal fossils in the Kuk-Karauk Formation. A preliminary identification of these fossil fragments (oral communication, L. E. Popov) suggests that they might be *Obolidae* brachiopods which appeared in Middle Cambrian. If confirmed, this finding will undermine the suggestion about Late Ediacaran age of Kuk-Karauk and Zigan formations.

Puchkov (2012a) argue that correlation with more precisely dated successions in Middle Urals suggest an age of ca. 560 Ma for the lower part of the Basu Formation and the age of ca. 570 Ma for the upper part of the Bakeevo Formation. Additionally, the diamictite in the lower part of the Bakeevo Formation might be related to the ca. 580 Ma Gaskiers glaciations (Bowring et al. 2003; Narbonne 2008). In summary, Puchkov (2012a) suggests that the age of the Bakeevo Formation is between 585 and 570 Ma, or, more conservatively, between 600 and 570 Ma.

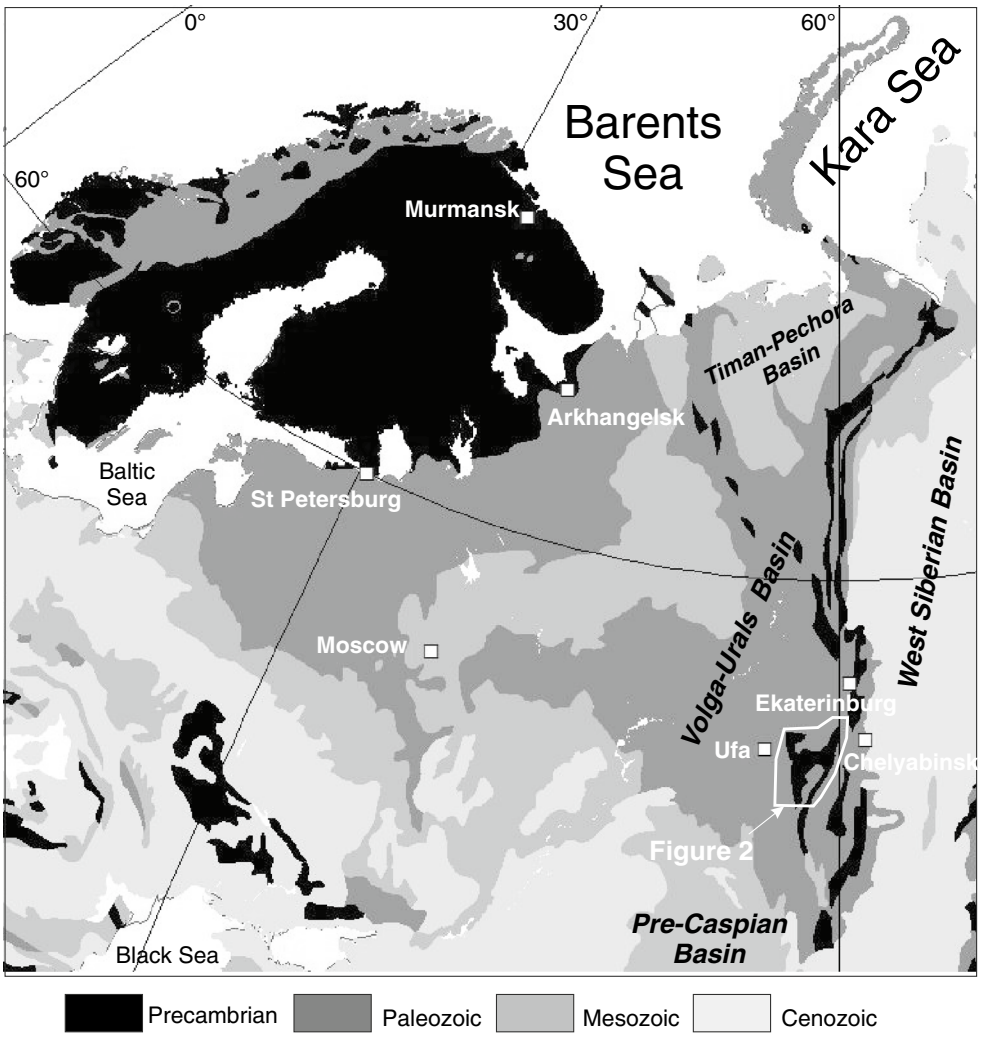


Fig. 1 Simplified geology of Baltica

The Kurgashlya Formation in the southern BMA is the correlative to the Bakeevo Formation. It contains conglomerates in the lower part and unconformably overlies the Upper Riphean Krivaya Luka Formation (Fig. 2). The latter is intruded by the yet undated dolerite body, and dolerite fragments are found in the Kurgashlya Formation. The BMA strata have been affected by Paleozoic deformation related to a collision with and island arc in the Famennian–Tournaisian followed by a continent–continent collision in the Middle Carboniferous–Late Permian (Puchkov 2010).

In 2000–2007, we collected oriented block samples in two localities: (1) near Bainazarovo village (Krivaya Luka loop of Belaya river), southern BMA (Fig. 2): 66 samples from the Cryogenian Krivaya Luka Formation) and 139 samples from the Ediacaran Kurgashlya Formation including 32 samples from the fine-grained basal matrix of conglomerates in the lower part of the formation; (2) near Ust-Katav town in the northern BMA: 62 samples from the Ediacaran Bakeevo Formation and ten samples from the sedimentary pebbles of the basal conglomerates of this formation.

In the Bainazarovo area (Fig. 2), we targeted quartz siltstones and sandstones of the Cryogenian Krivaya Luka Formation (>300 m thick) and tillite-like conglomerate and red siltstones/mudstones of the Ediacaran Kurgashlya Formation (~190 m thick). Oriented samples from the Krivaya Luka and Kurgashlya Formations were collected along a few isolated outcrops at Krivaya Luka loop of Belaya River. All strata are tilted southward by 30°–35° (Krivaya Luka Formation: 35° toward 185°; Kurgashlya Formation: 30° toward 192°). The two formations are separated by angular unconformity.

In the Ust-Katav area (Fig. 2), greenish and gray sandstones, siltstones and mudstones of the Bakeevo Formation overlie Upper Riphean (Cryogenian) stromatolitic limestones of the Uk Formation with angular unconformity and basal conglomerates. Oriented samples were collected from ~140 m thick along railway track near Ust-Katav town. The Bakeevo strata dip gently (18°) to the west (268°). In the upper part of this section, cherry-red and bluish green siltstones and mudstones of the Zigan Formation are

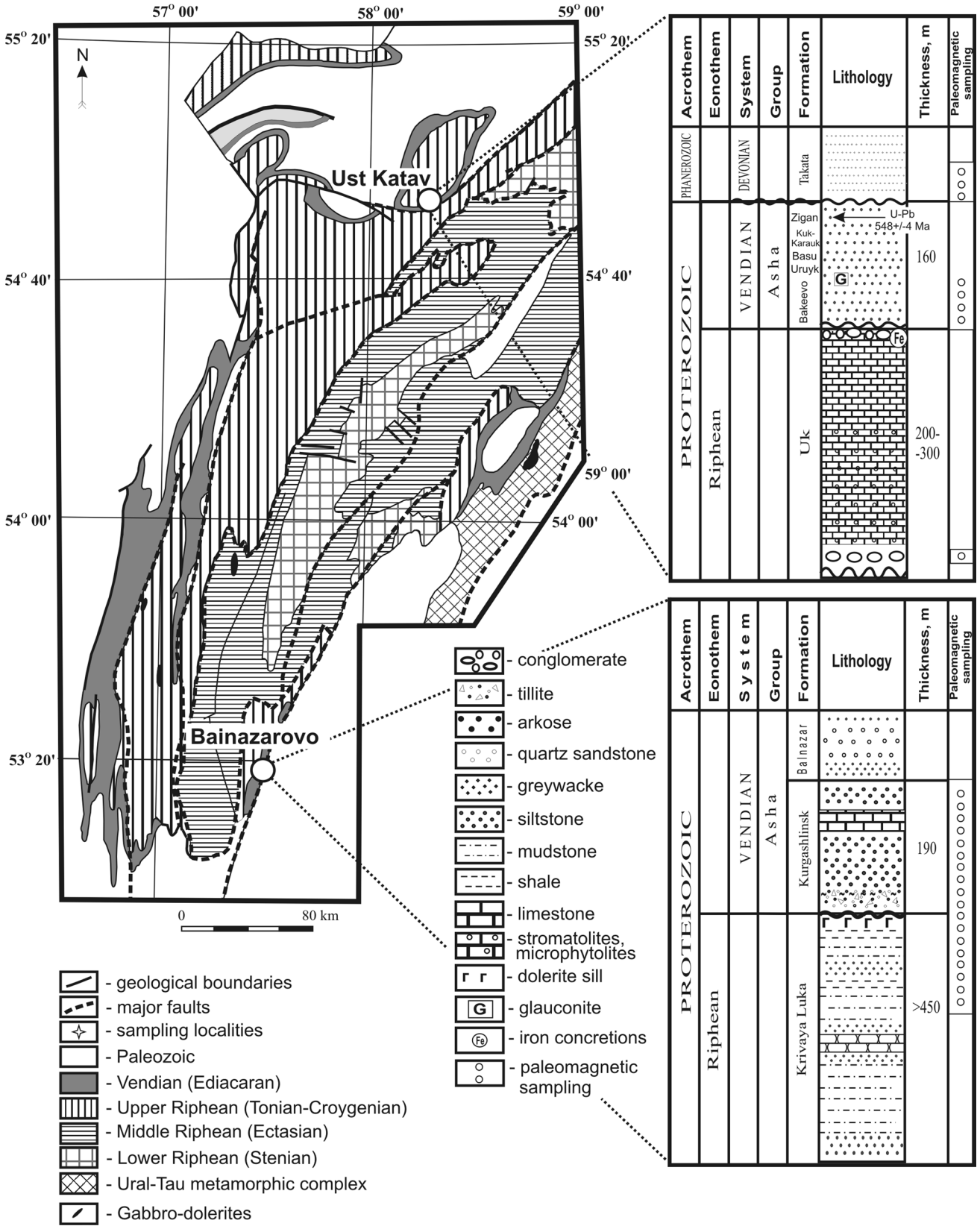


Fig. 2 Map of BMA and stratigraphy of the studied successions

unconformably overlain by Devonian quartz sandstones of the Takata Formation. Contacts between Vendian formations are not exposed.

Altogether, we collected 277 oriented hand samples from two localities. A magnetic compass was used to determine their orientations.

Analytical techniques

In the laboratory, two to four 2 cm cubic specimens were cut from each oriented block. Samples were analyzed in the Petromagnetic Laboratory of the Moscow State University, in Paleomagnetic Laboratory of VSEGEI (St Petersburg, Russia) and in the Tectonics Special Research Centre at the University of Western Australia. Magnetic remanence was measured using 2G-755R cryogenic magnetometer and JR-5A and JR-6 spinner magnetometers (Geofyzika, Brno, Czech Republic). Remanence composition was determined by detailed stepwise thermal demagnetization (≤20 steps, to 600 °C), using a MMTD2 furnaces manufactured by *Magnetic Measurements* (~10 nT residual field). Magnetization vectors were isolated using principal component analysis (Kirschvink 1980).

Paleomagnetic analysis

The natural remanent magnetization (NRM) of the studied sediments ranges from 5 to 100 mA/m, and their magnetic susceptibility from 50 to 200 $\times$ 10^{-6} SI units.

Bainazarovo location, Krivaya Luka Formation

During thermal and alternative field (AF) demagnetizations, after the removal of a low-stability, randomly oriented overprint, most samples exhibit two remanence components. The first, middle-temperature component (MT) has been isolated at 300–560 °C or below 25 mT (Fig. 3). This component has a SW shallow down direction (Fig. 4a; Table 1, entry 1). The high-temperature and high-coercive (>570 °C, >40 mT) component (KL) is bipolar with a NNW downward (SSE upward) direction (Figs. 3, 4b; Table 2, entries 2–4). The distribution of unblocking temperatures suggests that this component is carried both by magnetite (Fig. 3a) and by hematite (Fig. 3c). The reversal test of McFadden and McElhinny (1990) is positive (Rc, $\Delta = 3.8°$, $\Delta c = 14.4°$).

Bainazarovo location, Kurgashlya Formation

Thermal and AF demagnetizations isolated either two or one remanence components (Fig. 5). The middle-temperature component (MT) is similar to that in the Krivaya Luka Formation (Table 1, entry 5; Figs. 4a, 5a, b, 6a). The high-temperature (560–600 °C) and high-coercive (up to 90 mT) component (KG) is bipolar with a NW–SE declination and medium inclination and is carried mostly by the magnetite (Table 2; Figs. 5, 6b). The lower part of the Kurgashlya Formation is mostly represented by conglomerates with a fine-grained basic matrix. Demagnetization of samples taken from this matrix reveal the KG component exclusively with a "reverse" (SE, upward) polarity, while the upper part of the formation comprised by red sandstones and siltstones is magnetized "normally" (NE, downward). We use quotation marks for "normal" and "reverse" because of the polarity ambiguity of Precambrian paleomagnetic data. This magnetostratigraphic pattern provides an indirect evidence for the primary nature of the KG component, because such pattern is unlikely to be caused by a bipolar secondary remagnetization. The reversal test of McFadden and McElhinny (1990) is positive (Rb, $\Delta = 5.9°$, $\Delta c = 9.8°$).

Ust-Katav location, Bakeevo Formation

In most cases, apart from the unstable low-temperature component, both thermal and AF demagnetization of the samples from Bakeevo Formation isolated a single, stable, bipolar remanence component (BK) directed moderately downward to NW or upward to SE (Table 2, entries 2–4; Figs. 7, 8b). The unblocking temperatures are up to 680 °C (Fig. 7a, c). The reversal test of McFadden and McElhinny (1990) is positive (Rc, $\Delta = 10.1°$, $\Delta c = 11.2°$). Few AF demagnetizations also show a high stability of this component (Fig. 7b). This implies that both magnetite and hematite carry the remanence of similar age. Twelve samples from the upper part of the formation also exhibit a medium-temperature (MT) unipolar W-directed shallow upward component (Table 1, entry 1; Figs. 7c, 8a).

Ust-Katav location, basal conglomerates of the Bakeevo Formation

Sedimentary pebbles of the conglomerate have been sampled very close to the outcrops of the sandstones of the Bakeevo Formation. These rocks mostly carry a single remanence component (Fig. 9a, b), which is randomly oriented (Fig. 9c). The Rayleigh statistics $r = R/n = 0.196$ compared with the critical value $r_c = 0.503$ for ten pebbles at 95 % confidence level (Mardia 1972), so the conglomerate test is positive. This is not a conclusive proof for the primary origin of the BK component because the tested pebbles underlie the sandstones of the Bakeevo Formation. However, the closeness of locations implies that the probability of a later remagnetization event is low.

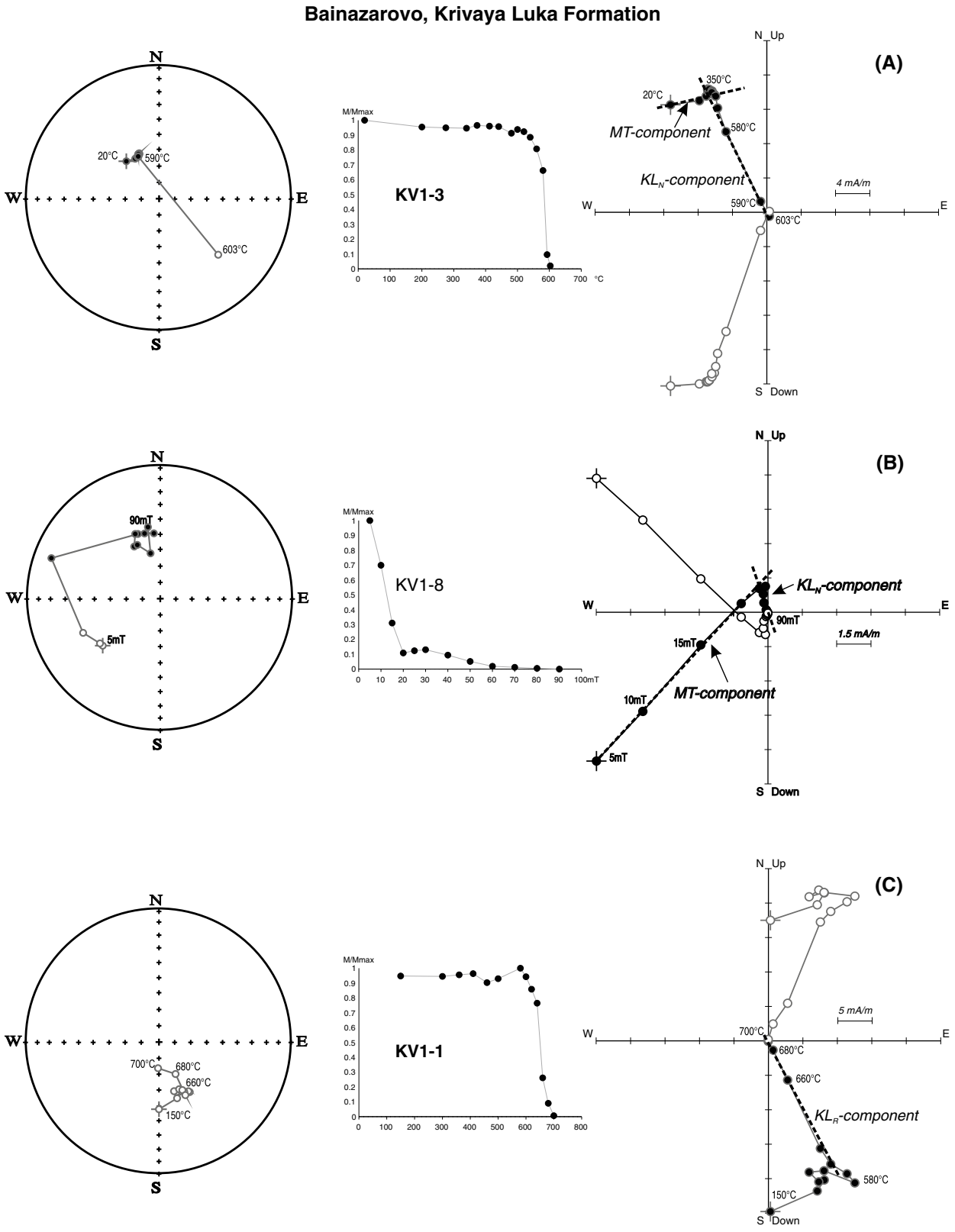


Fig. 3 Thermal and AF demagnetizations, Krivaya Luka Formation. In orthogonal plots, *open* (*closed*) *symbols* show magnetization vector endpoints in the *vertical* (*horizontal*) *plane*; *curves* show changes in intensity during demagnetization. *Stereoplots* (equidistant projection) show upwards (*downwards*) pointing paleomagnetic directions with *open* (*closed*) *symbols*. **a** Sandstones, thermal demagnetization; **b** sandstones, AF demagnetization; **c** sandstones, thermal demagnetization

Ages of remanences

Paleopoles calculated from MT components (tilt corrected) in Krivaya Luka and Kurgashlya Formations lie close to the Late Carboniferous–Early Permian part of the Baltican APWP (Fig. 10; Smethurst et al. 1998). They are probably caused by a remagnetization related to the continent–continent collision (Puchkov 2010). These processes did not affect the strata in the Ust-Katav area, which are more distant from the collisional front. However, the MT pole of the Bakeevo Formation in Ust-Katav area falls on the Devonian part of the Baltican APWP (Fig. 10) and might be related to the Devonian magmatism or the Late Devonian arc–continent collision (Puchkov 2012b).

The anti-polarity of the high-temperature remanences of the lower and upper parts of the Kurgashlya Formation supports the primary nature of this component (KG_{N+R}, Table 2). This evidence, however, is not conclusive because of a lithological difference between these two parts of the formation discussed above. The high-temperature component BK in the Bakeevo Formation is also bipolar, but it does not show a clear magnetostratigraphic pattern. However, the random remanence directions in pebbles of the basal conglomerate suggest the absence of later strong remagnetization events in the Ust-Katav area. Paleopoles calculated from KG and BK components lie far away from any Phanerozoic poles, but close to other Ediacaran poles (Fig. 10; Table 3). BK and KG poles are close to each other, but their ovals of confidence do not overlap (Fig. 10). As the paleomagnetic declinations of the BK and KG ChRMs are almost identical (316° vs. 313°, Tables 2, 3), but their inclinations are significantly different (36° vs. 52°), it is unlikely that the difference is caused by the later local rotations between Ust-Katav and Krivaya Luka areas. Lubnina (2009) and Levashova et al. (2013) carried out a special study and conclude that the Late Ediacaran or later local rotations in the western part of the Southern Urals were unlikely. So, it is more likely that Bakeevo and Kurgashlya Formations are similar in age, but not exactly coeval, and this is supported by the difference in their magnetostratigraphic pattern. We agree with Levashova et al. (2013) that paleopoles from Ediacaran sedimentary rocks of Southern Urals are representative of cratonic Baltica. On the other hand, BK and KG poles are quite distinct from the recently published 548 Ma paleopole of the Zigan Formation—the topmost unit in the Ediacaran succession of the western BMA (Zig in Table 3; Fig. 10; Levashova et al. 2013). If all three (BK, KG and Zig) poles are primary, this difference is most likely explained by the drift of Baltica during that time. This interpretation is also supported by positions of paleopoles from the 555–550 Ma sedimentary rocks in the Arkhangelsk area, which are slightly older than the Zigan Formation, but probably younger than Bakeevo and Kurgashlya formations (Win, Zol1, Zol2 and Ver in Table 3; Fig. 10). These poles, though dispersed, lie between Zig and BK poles along the smooth and relatively short line, which we consider as an approximation of the Late Ediacaran APWP for Baltica (dashed line in Fig. 10). The paleopole from the Vendian (Ediacaran) Podolian strata (Iosifidi et al. 2005) also lies close to this line, but these data were criticized by Levashova et al. (2013) due to difficulties in distinguishing between various overprints.

The paleopole KL calculated for the high-temperature remanence component of the Krivaya Luka Formation, supported by the reversal test, lies relatively close to BK and KG poles (Table 2; Fig. 10). The age of the Krivaya

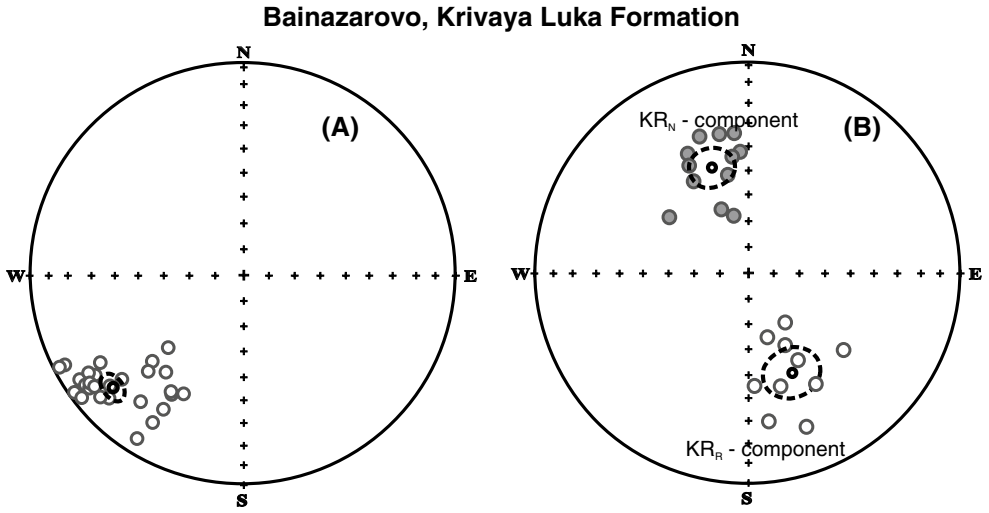


Fig. 4 Stereoplots of the sample mean directions of the remanence components in the sandstones of Krivaya Luka Formation: **a** Mid-temperature component (MT); **b** high-temperature component of normal (KR_N) and reverse (KR_R) polarities. *Projection* and *symbols* as in Fig. 3

Table 1 Paleomagnetic directions and poles of the mid-temperature components

#	Component	Formation	Rocktype	*N*	In situ				Tilt corrected				Plat (°N)	Plong (°E)	Dp (°)	Dm (°)
					Dec (°)	Inc (°)	*k*	α_{95} (°)	Dec (°)	Inc (°)	*k*	α_{95} (°)				
Krivaya Luka and Kurgashlya Formations, Bainazarovo village, South Urals (53.32°N, 57.53°E)																
1	MT	Krivaya Luka	Sandstone	31	229.1	4.8	82.9	5.1	232.6	−20.0	82.9	5.1	30.0	173.0	2.8	5.3
2	MT	Kurgashlya	Conglomerate*	10	223.1	−4.9	18.6	11.5	228.1	−30.9	18.6	11.5	37.8	173.1	7.2	12.8
3	MT		Siltstones, sandstone	25	221.8	−11.0	42.6	4.5	228.7	−37.1	42.6	4.5	40.7	169.5	3.1	5.3
Bakeevo Formation, town of Ust-Katav, South Urals (54.94ºN, 58.16ºE)																
4	MT	Bakeevo	Siltstones, sandstone	12	281.5	−20.5	14.4	11.9	278.0	−29.9	14.4	11.9	8.6	312.5	7.3	13.2

N = number of independently oriented blocks used; Dec, Inc = site mean declination, inclination; *k* = best estimate of the precision parameter of Fisher (1953); α_{95} = the semi-angle of the 95 % cone of confidence; Plat, Plong = latitude, longitude of the paleopole; Dp, Dm = the semi-axes of the cone of confidence about the pole at the 95 % probability level

* Only fine-grained matrix has been used for the sampling

Table 2 Paleomagnetic directions and poles of the high-temperature components

#	Component	Formation	Rocktype	*N*	In situ				Tilt corrected				Plat (°N)	Plong (°E)	Dp (°)	Dm (°)
					Dec (°)	Inc (°)	*k*	α_{95} (°)	Dec (°)	Inc (°)	*k*	α_{95} (°)				
Krivaya Luka and Kurgashlya Formations, Bainazarovo village, South Urals (53.32°N, 57.53°E)																
1	KL_N	Krivaya Luka	Sandstone	10	165.6	−13.9	22.3	10.5	157.2	−46.2	22.3	10.5	59.2	279.7	8.6	13.5
2	KL_R		Sandstone	12	347.8	13.6	29.2	8.2	340.3	46.5	29.2	8.2	60.6	275.0	6.8	10.5
3	KL		Sandstone	22	346.8	13.8	26.8	6.1	**338.9**	**46.4**	**26.8**	**6.1**	**58.6**	**275.8**	**4.8**	**7.7**
4	KG_N	Kurgashlya	Siltstones, sandstone	27	336.0	34.5	20.7	6.3	312.6	54.2	20.7	6.3	52.0	317.8	6.2	8.9
5	KG_R		Conglomerate*	17	152.9	−29.8	24.1	7.4	133.0	−48.8	22.9	14.3	49.2	310.3	6.4	9.7
6	KG_{N+R}		Conglomerate*, siltstones, sandstones	44	334.9	32.4	21.8	4.7	**313.2**	**51.9**	**21.8**	**4.7**	**50.9**	**314.5**	**4.4**	**6.4**
Bakeevo Formation, town of Ust-Katav, South Urals (54.94°N, 58.16ºE)																
7	BK_N	Bakeevo	Siltstones, sandstone	27	311.4	51.6	11.6	8.5	312.7	39.7	11.6	8.5	51.7	316.4	6.2	8.9
8	BK_R		Siltstones, sandstone	21	141.3	−43.9	17.9	7.7	140.8	−31.9	17.9	7.7	48.7	129.2	6.4	9.7
9	BK_{N+R}		Siltstones, sandstone	48	316.2	48.3	13.3	5.9	**316.5**	**36.3**	**13.3**	**5.9**	**42.3**	**299.1**	**4.0**	**6.9**

The significance in bold is used for further reconstruction

See footnotes to Table 1

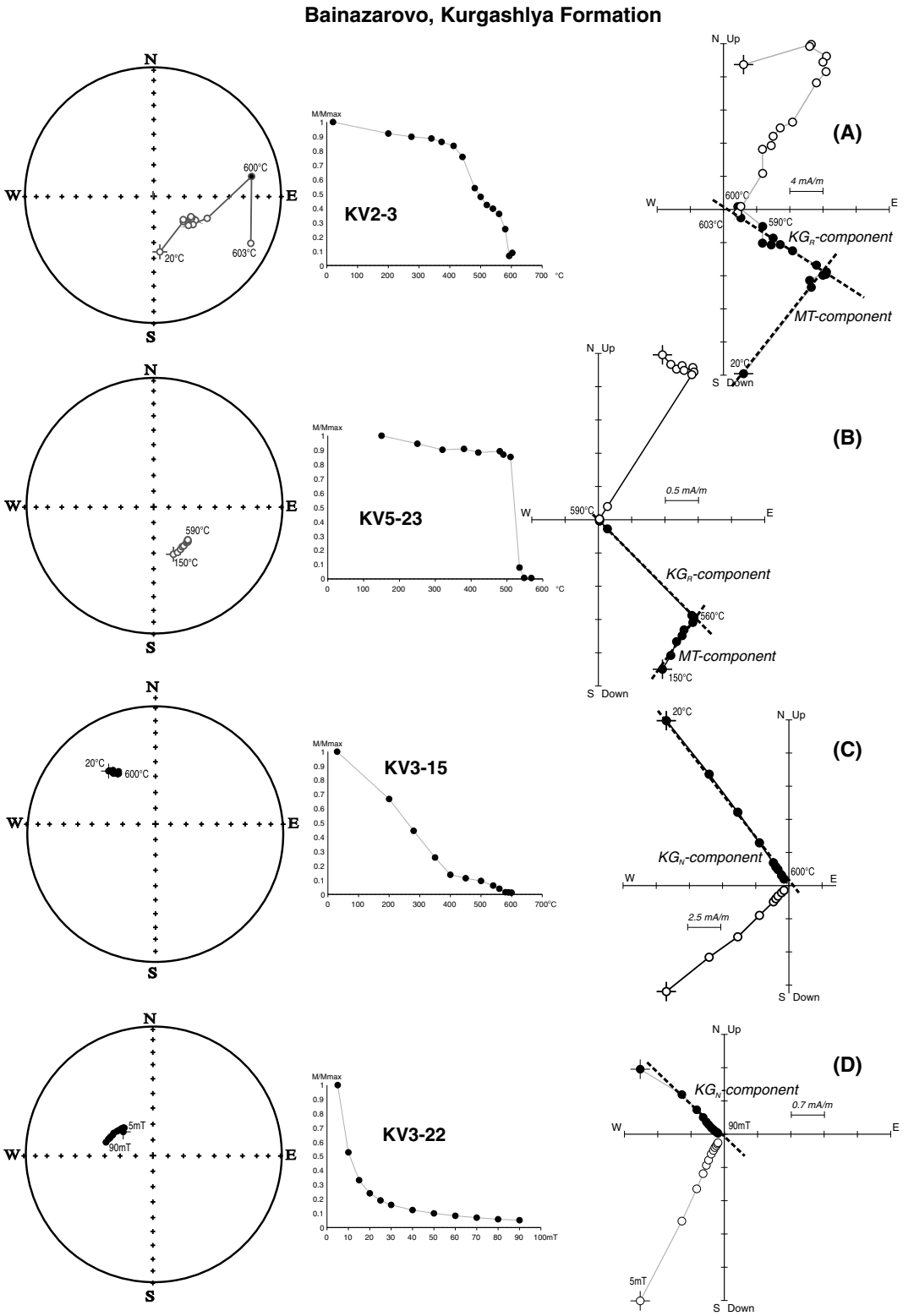


Fig. 5 Thermal and AF demagnetizations, Kurgashlya Formation: **a** conglomerate matrix, thermal demagnetization; **b** conglomerate matrix, thermal demagnetization; **c** siltstones and sandstones, thermal demagnetization; **d** siltstones and sandstones, AF demagnetization. *Projection* and *symbols* as in Fig. 3

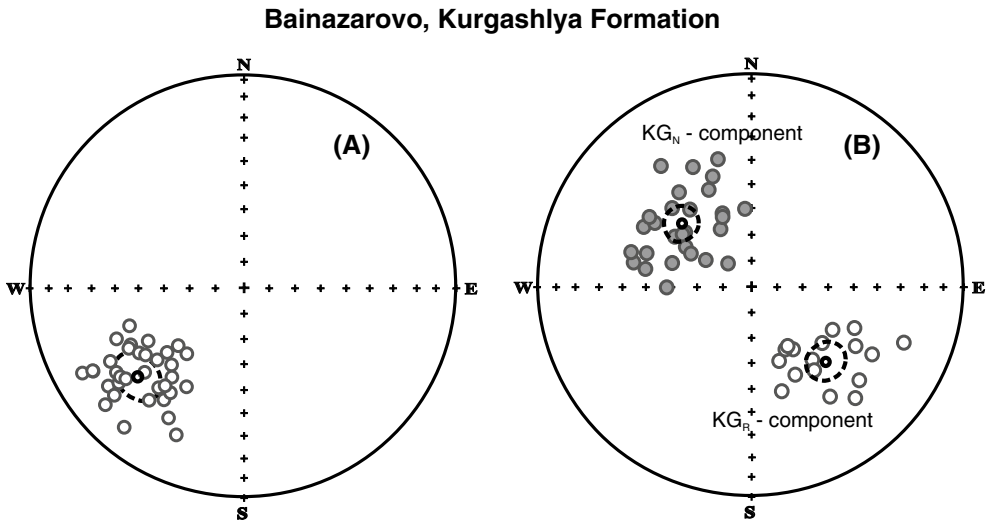


Fig. 6 Stereoplot of the sample mean directions of remanence components in the Kurgashlya Formation: **a** mid-temperature component (MT); **b** high-temperature component (KG). *Projection* and *symbols* as in Fig. 3

Luka Formation is poorly known. It must be older than the 710 ± 10 Ma porphyries in the overlying Arsha Group and is stratigraphically constrained as Late Riphean (Tonian–Cryogenian) in age (Puchkov 2010). Our KL pole is significantly different from (1,000–850 Ma) Baltican poles (see Elming et al. in press for an overview). Unfortunately, no well-dated reliable 850–700 Ma Baltican poles are published yet. Shipunov and Chumakov (1991) reported a pole (13°N, 195°E) from the Neoproterozoic red beds in the northern part of the Kola Peninsula (Russia) supported by a positive fold test, but their age is poorly dated between 790 and 600 Ma. In any case, this pole is also far away from our KL pole. On the other hand, the KL pole is close to BK and KG poles (Fig. 10), so there is a possibility that the KL component represents an Ediacaran overprint and we excluded this component from the further discussion.

Ediacaran paleogeography in the eastern Iapetus' realm

It is generally accepted that the Iapetus Ocean opened after ca. 600 Ma during the breakup between Laurentia, Baltica and probably Amazonia (e.g., Pisarevsky et al. 2008 and references therein). However, all attempts to restore the details of this process are hampered by the controversy about some Ediacaran paleopoles from Laurentia, illustrated in Fig. 11a and in Table 3. There are several coeval pairs of poles with ages between ca. 585 and 565 Ma, which are located very far from each other. Most of these poles are supported by field tests, but neither of them is conclusively primary. Contrasting paleomagnetic directions have been retrieved from the same rocks, i.e., Catoctin Volcanics (Meert et al. 1994) and Baie des Moutons Syenite (McCausland et al. 2011). Halls and Hamilton (2012) demonstrated an impressive example from precisely dated Grenville dykes where the steep magnetization was followed by the shallow one in <4 m.y., which is probably too quick even if the inertial interchange true polar wander (IITPW) hypothesis is suggested (see below). Unfortunately, the details of this study are not yet published. Buchan et al. (2011) also discussed this controversy. Importantly, these two contrasting groups of paleomagnetic data imply contrasting paleopositions of Laurentia—high-latitude *vs* low-latitude. This controversy has been widely discussed (e.g., Meert and Van der Voo 2001; Pisarevsky et al. 2000, 2001; Evans 2003; Hodych et al. 2004; Cawood and Pisarevsky 2006; Pisarevsky et al. 2008; Abrajevitch and Van der Voo 2010; McCausland et al. 2011; Meert 2014). A general consensus has emerged from this body of the literature that there is no way to reconcile these two sets of data in the frame of traditional plate tectonics because the calculated speed of Laurentia has to be unreasonably high. Three suggestions have been proposed to explain this paradox: (1) one (or possibly both) of the groups of Laurentian paleomagnetic data does not represent the primary remanence; (2) both remanences are primary, but there were one or several episodes of the IITPW between ca. 585 and 565 Ma (e.g., Evans 2003); (3) both remanences are primary, but there was a significant disturbance in the geomagnetic dynamo between ca. 585 and 565 Ma, which caused an unusual configuration of the geomagnetic field (equatorial dipole) (Abrajevitch and Van der Voo 2010). Importantly, even if we accept any of these suggestions, it would not affect the plate tectonics scenario of the opening of Iapetus—neither IITPW nor the anomalous behavior of

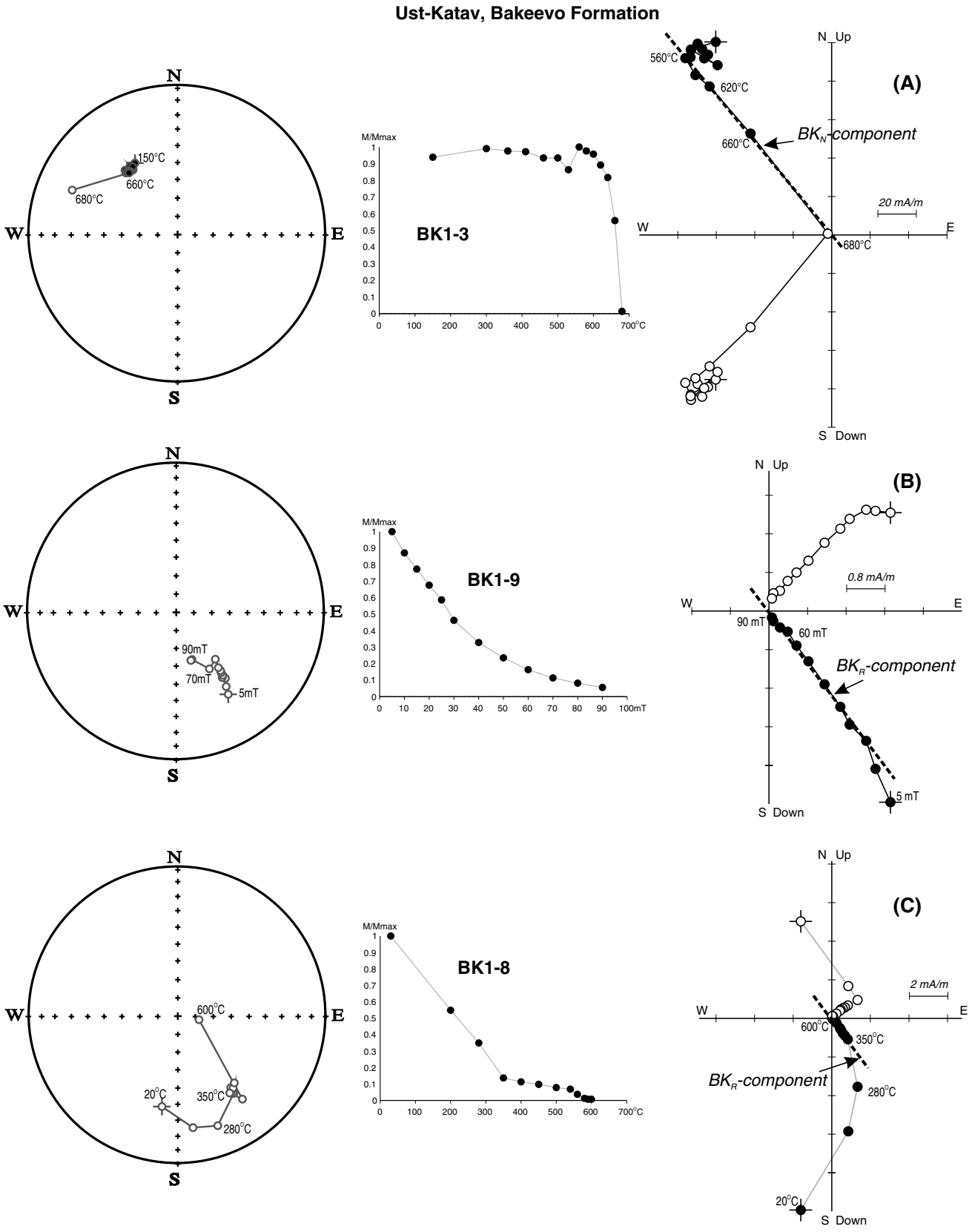


Fig. 7 Thermal and AF demagnetizations, Bakeevo Formation: **a** siltstones and sandstones, thermal demagnetization; **b** siltstones and sandstones, AF demagnetization; **c** siltstones and sandstones, thermal demagnetization, reversal polarity (BK_R). *Projection* and *symbols* as in Fig. 3

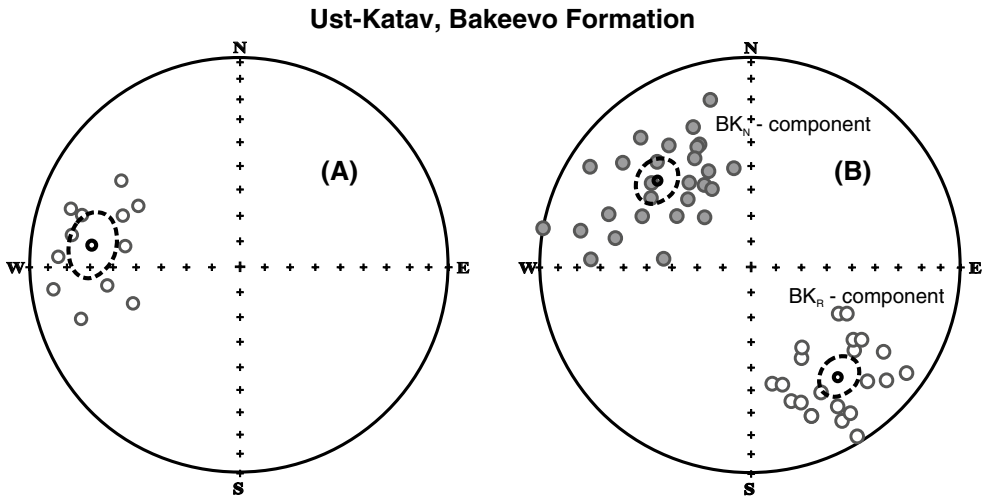


Fig. 8 Stereoplot of the sample mean directions of the remanence components in the Bakeevo Formation: **a** mid-temperature component in siltstones and sandstones (MT); **b** high-temperature component normal (BK_N) and reversal (BK_R) polarities. *Projection* and *symbols* as in Fig. 3

the geomagnetic field apparently should cause any significant changes in the plate tectonic kinematics, so the *relative* movements of Laurentia and Baltica can be restored by traditional methods, if the controversial paleopoles are ignored. For our reconstructions, we applied the usual approach of minimizing the plate movements.

After excluding the controversial paleopoles, we are left with only three Laurentian Ediacaran poles available for the paleogeographic reconstructions: the 615 ± 2 Ma Long Range dykes Laurentian pole supported by the contact test (Table 3; Fig. 11b; Murthy et al. 1992), the slightly younger 605 ± 10 Ma Cloud Mountain Basalts pole of Deutsch and Rao (1977) (Table 3; Fig. 11b) and the 550 ± 3 Ma Skinner Cove pole (Table 3; Fig. 11b; McCausland and Hodych 1998). Levashova et al. (2013) additionally considered the Johnnie Formation paleopole (JF in Table 3; Fig. 11). The age of this pole is imprecisely estimated by the stable carbon isotope intercontinental correlation at 570 ± 10 Ma (Hodych et al. 2004 and references therein), so it is possible that the remanence has been acquired after the controversial 585–565 Ma time interval. The position is close to the shortest line between LR and SC poles (Fig. 11b).

The 616 ± 3 Ma (U–Pb, baddeleyite; Bingen et al. 1998)/609 ± 10 Ma ($^{40}Ar/^{39}Ar$, biotite) Egersund Dykes Baltican pole (Table 3; Walderhaug et al. 2007) is coeval to the Laurentian LR pole. This pole lies close to the Ordovician part of the Baltican APWP (Fig. 10), but the robust contact test suggests the remanence is primary, not a later overprint (Walderhaug et al. 2007). A recent study of the Nyborg Formation in Finnmark (Norway) retrieved a relatively close paleopole at 21°N, 79°E, $A_{95} = 21°$ (Hovland 2012; Walderhaug, personal communications). The age of the Nyborg Formation is not well constrained, but the formation is sandwiched between the Smalfjord Tillite (correlative to the 635 Ma Marinoan Tillite) and the Mortensnes Tillite (correlative to 580 Ma Gaskiers Tillite) (e.g., Knoll et al. 2006). The angular difference between Nyborg and Egersund poles is ca. 30° and their ovals of confidence overlap, which is compatible with "normal" plate kinematics. No other reliable pre-585 Ma Ediacaran paleopoles from Laurentia and Baltica have been reported, so at present, we have no reason to support either an IITPW event or significant distortions of the geomagnetic field at that time. There are five well-dated latest Ediacaran poles from Baltica (Table 3; Fig. 10), which have been discussed in previous section. Ages of our new BK and KG poles and of the Pod pole (Iosifidi et al. 2005) are not well constrained, so we cannot be sure that these remanences have been acquired after the "controversial" 585–565 Ma time period. However, their closeness to each other and to high-quality 555–550 Ma poles (Fig. 10) suggests that they probably have; otherwise, we should rather expect larger scatter of these poles as in the case of controversial Laurentian poles (Fig. 11a). Previously reported paleopoles from ca. 590 to 580 Fen and Alnø carbonatite complexes in Baltica (Meert et al. 1998, 2007) have been recently classified as "low-reliability" by Meert (2014), so we shall not consider them here. Hence, the ca. 600–570 Ma time interval is not paleomagnetically constrained for Baltica.

In Fig. 12, we show a set of paleogeographic reconstructions of Laurentia and Baltica between 615 and 550 Ma. We used chosen paleomagnetic poles and interpolation for paleomagnetically unconstrained time intervals. Euler rotation parameters for each reconstruction are shown in Table 4. Following the previous studies (e.g., St Seymour and Kumarapeli 1995; Ernst and Buchan 1997; Puffer

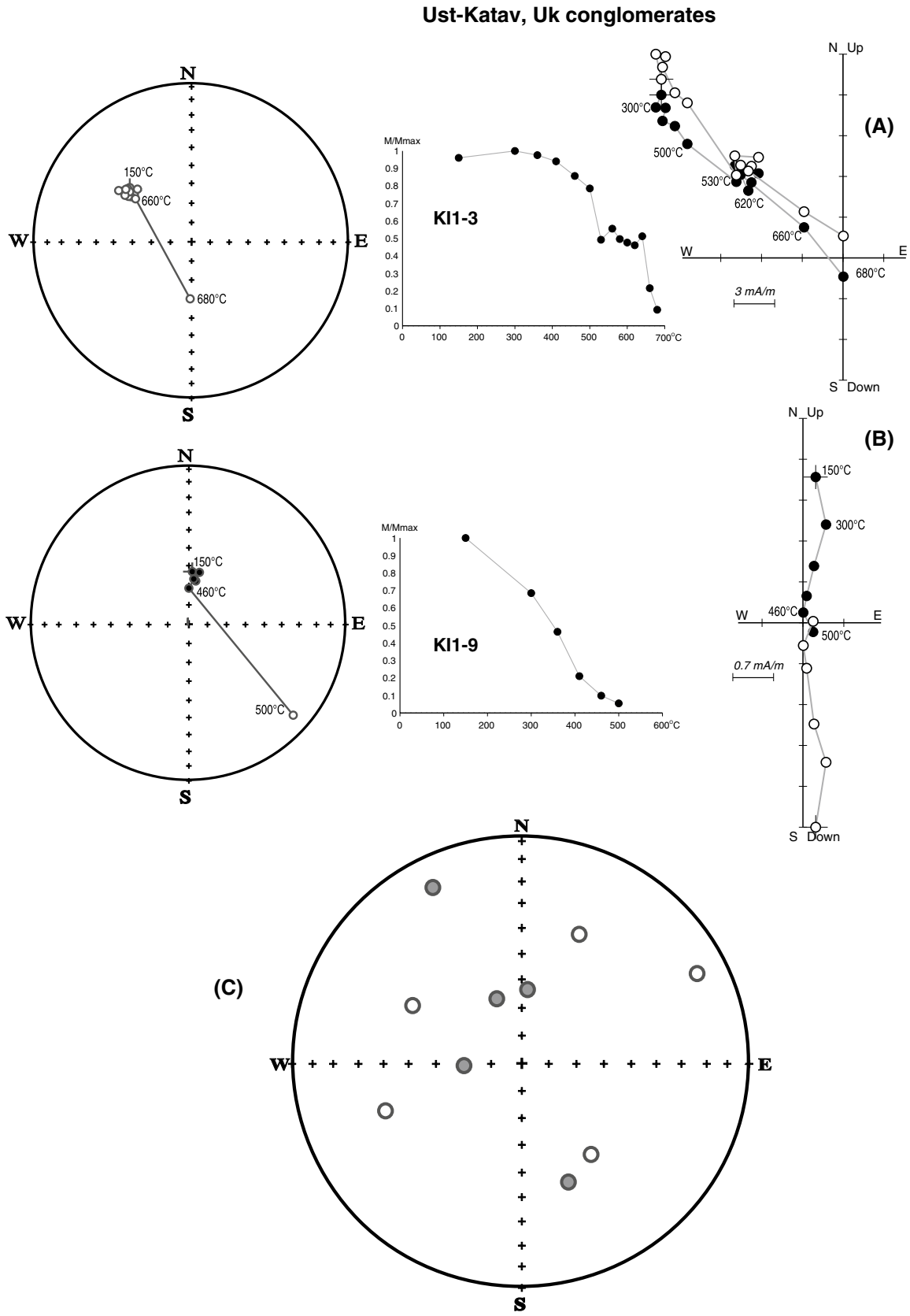


Fig. 9 Stereoplots and examples of demagnetization behavior for the basal conglomerates of the Bakeevo Formation: **a**, **b** conglomerates, thermal demagnetization; **c** stereoplot of the sample mean directions of ChRM in the Uk conglomerates. *Projection* and *symbols* as in Fig. 3

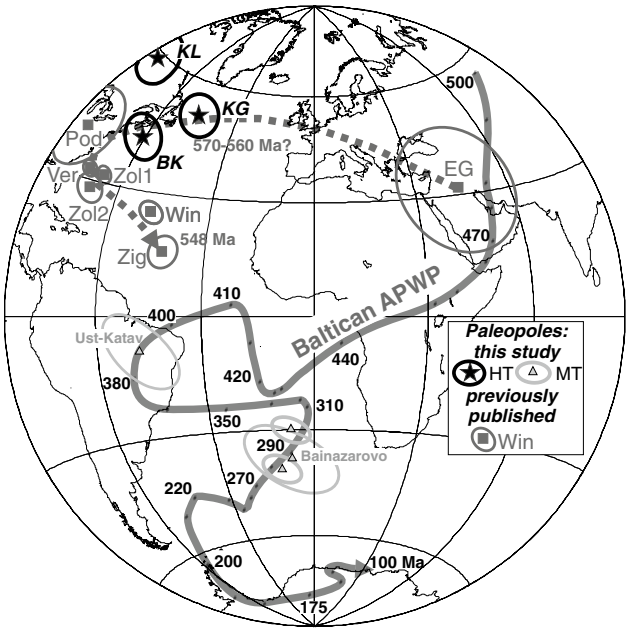


Fig. 10 The 500–100 Ma apparent polar wander path (APWP) for Baltica (after Smethurst et al. 1998) and paleopoles calculated in this study. Abbreviations are as in Table 3. *HT,* high-temperature component (*black stars*); *MT,* mid-temperature component (*gray triangles*). Dashed line = the suggested Ediacaran APWP for Baltica

2002), Pisarevsky et al. (2008) argue that the breakup of the joint Laurentia–Baltica–Amazonia continent has been initiated by a mantle plume. The first magmatism related to this plume is represented by 615 Ma Long Range and Egersund dyke swarms. The rift-to-drift transition was successfully completed at ca. 590 Ma and led to the opening of the eastern Iapetus Ocean and the Tornquist Sea in Ediacaran times (Bingen et al. 1998; Greiling et al. 1999; Siedlecka et al. 2004). The eastward movement of the Baltican plate is coeval with the initiation of subduction somewhere in the Mirovoi Ocean. After ca. 570 Ma, subduction zones approached Timanian and Uralian continental margins of Baltica and the passive margin has been inverted into the active margin, which caused the development of foreland basins (Roberts and Siedleska 2002; Puchkov 2010).

Conclusions

Our paleomagnetic study of Bakeevo and Kurgashlya formations of the Southern Urals reveals stable high-temperature remanences that are interpreted to be primary. Two new paleomagnetic poles together with well-dated high-quality group of recently published 555–550 Ma Baltican poles provide an opportunity to construct a latest Ediacaran

Table 3 Ediacaran (Vendian) paleomagnetic poles, Baltica and Laurentia

Object	Pole		dp/dm	*Q*	Age (Ma)	References
	(°N)	(°E)	(°)			
Baltica						
Egersund dykes, Norway (EG)	31	44	15/17	6	616 ± 3	Walderhaug et al. (2007) and Bingen et al. (1998)
					609 ± 10	
Kurgashlya formation (KG)	51	315	4/6	5	570–560	This study
Bakeevo Formation (BK)	42	299	4/7	5	570–560	This study
Podolian sediments, Ukraine (Pod)	40	277	8/8	4	580–550	Iosifidi et al. (2005)
Winter coast sediments, Russia (Win)	25	312	2/4	6	555 ± 1	Popov et al. (2002) and Martin et al. (2000)
Zolotitsa sediments I, Russia (Zol1)	32	293	2/3	6	550 ± 5	Popov et al. (2005)
Verkhotina sediments, Russia (Ver)	32	287	2/3	5	550 ± 1	Popov et al. (2005)
Zolotitsa sediments II, Russia (Zol2)	28	290	4/4	6	550 ± 5	Iglesia Llanos et al. (2005)
Zigan formation, Russia (Zig)	16	318	4/4	6	548 ± 4	Levashova et al. (2013)
Laurentia						
Long Range dykes (5 dykes)* (LR)	19	355	15/21	5	615 ± 2	Murthy et al. (1992) and Kamo and Gower (1994)
Cloud Mountain Basalt (CM)	−5	352	2/4	3	605 ± 10	Deutsch and Rao (1977) and Stukas and Reynolds (1974)
Baie des Moutons Syenite A (BdM-A)	43	333	12/12	5	583 ± 3	McCausland et al. (2011)
Baie des Moutons Syenite B (BdM-B)	−34	322	11/22	5	583 ± 3	McCausland et al. (2011)
Callander complex (CC)	46	301	6/6	5	575 ± 5	Symons and Chiasson (1991)
Catoctin volcanics A (C-A)	43	308	9/9	5	564 ± 9	Meert et al. (1994) and Aleinikoff et al. (1995)
Catoctin volcanics B (C-B)	4	13	10/10	4	564 ± 9	Meert et al. (1994) and Aleinikoff et al. (1995)
Sept Iles intrusion A (SI-A)	−20	321	5/9	5	565 ± 4	Tanczyk et al. (1987) and Higgins and van Breemen (1998)
Johnnie formation (JF)	10	342	5/10	4	570 ± 10	Van Alstine and Gillett (1979) and Hodych et al. (2004)

dp/dm, the semi-axes of the cone of confidence about the pole at the 95 % probability level; *Q*, quality factor after Van der Voo (1990) and ranges from 0 to 7 with the later representing the highest quality data

* Recalculated by Hodych et al. (2004)

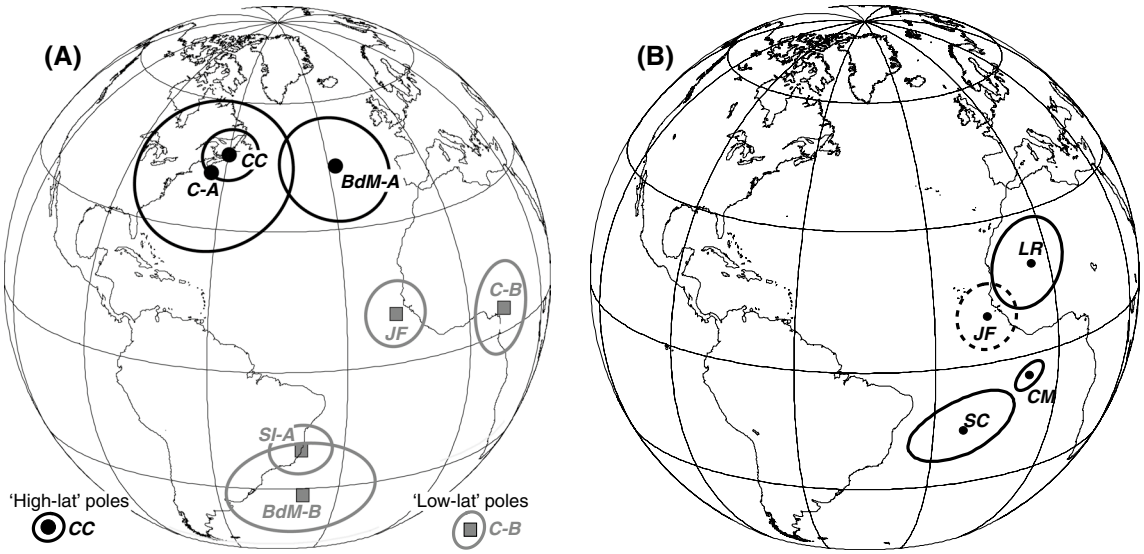


Fig. 11 Ediacaran Laurentian paleopoles: **a** controversy of 585–565 Ma poles; **b** Laurentian poles used for paleogeographic reconstructions

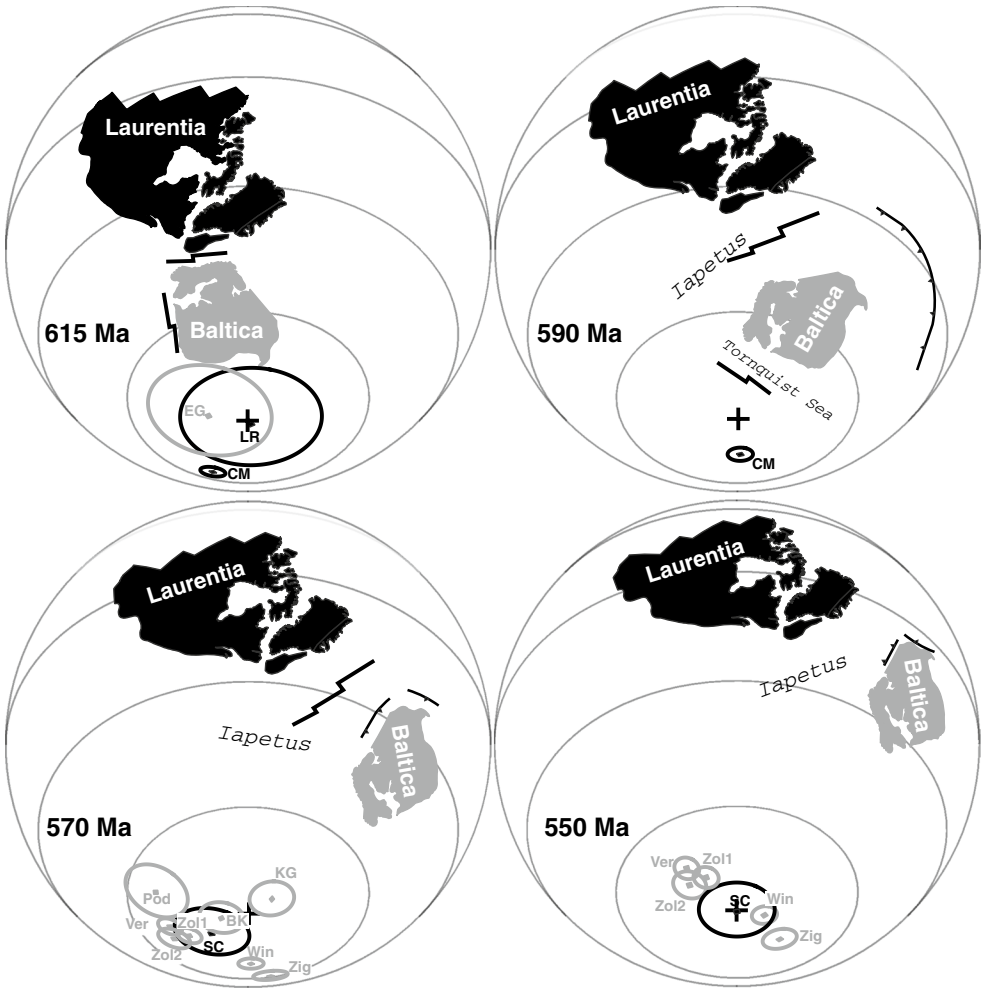


Fig. 12 Paleogeographic reconstructions of the breakup of Baltica and Laurentia. Euler rotation parameters—in Table 4

Table 4 Euler rotation parameters (to the absolute framework)

Continent	Pole		Angle
	(°N)	(°E)	(°)
615 Ma			
Laurentia	30.84	−155.19	−146.05
Baltica	28.74	−128.54	−183.60
590 Ma			
Laurentia	34.79	−155.55	−131.91
Baltica	25.85	−160.16	−163.39
570 Ma			
Laurentia	38.26	−155.89	−145.62
Baltica	14.19	178.94	−145.62
550 Ma			
Laurentia	42.14	−156.32	−109.80
Baltica	19.97	−179.56	−126.98

fragment of Baltican APWP. After selection of Ediacaran paleopoles from Laurentia and Baltica, we show a continuous 615–550 Ma paleogeographic model of the breakup of Laurentia and Baltica and the opening of the Iapetus Ocean.

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References

Abrajevitch A, Van der Voo R (2010) Incompatible Ediacaran paleomagnetic directions suggest an equatorial geomagnetic dipole hypothesis. Earth Planet Sci Lett 293:164–170

Aleinikoff JN, Zartman RE, Walter M, Rankin DW, Lyttle PT, Burton WC (1995) U–Pb ages of metarhyolites of the Catoctin and Mount Rogers formations, central and southern Appalachians: evidence for two phases of Iapetan rifting. Am J Sci 295:428–454

Bekker YuR (1992) The oldest Ediacaran biota in Urals. Transactions AN SSSR, Geological Series No 6: 16–24 (in Russian)

Bingen B, Demaiffe D, Van Breemen O (1998) The 616 Ma old Egersund basaltic dike swarm, SW Norway, and Late Neoproterozoic opening of the Iapetus Ocean. J Geol 106:565–574

Bogdanova SV, Bingen B, Gorbatschev R, Kheraskova TN, Kozlov VI, Puchkov VN, Volozh YuA (2008) The East European Craton (Baltica) before and during the assembly of Rodinia. Precambr Res 160:23–45

Bowring SA, Myrow P, Landing E, Ramezani J, Grotzinger JP (2003) Geochronological constraints on terminal Neoproterozoic events and the rise of metazoans. European Geophysical Union annual meeting, Nice 2003. Geophys Res Abstr 5:13219

Cawood PA, Pisarevsky SA (2006) Was Baltica right way up or upside down in the Neoproterozoic? J Geol Soc Lond 163:753–759

Cawood PA, McCausland PJA, Dunning GR (2001) Opening Iapetus: constraints from the Laurentian margin in Newfoundland. GSA Bull 113:443–453

Chumakov NM (2011) Late Proterozoic African glacial era. Stratigr Geol Correl 19(1):1–20

Collins AS, Pisarevsky SA (2005) Amalgamating Eastern Gondwana: the evolution of the Circum-Indian orogens. Earth-Sci Rev 71:229–270

Compston W, Sambridge MS, Reinfrank RF, Moczydlowska M, Vidal G, Claesson S (1995) Numerical ages of volcanic rocks and the earliest faunal zone within the late Precambrian of east Poland. J Geol Soc Lond 152:599–611

Cordani UG, D'Agrella-Filho MS, Brito-Neves BB, Trindade RIF (2003) Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments. Terra Nova 15(5):350–359

Dalziel IWD (1997) Neoproterozoic–Paleozoic geography and tectonics: review, hypothesis, environmental speculation. GSA Bull 109:16–42

Deutsch ER, Rao KV (1977) New palaeomagnetic evidence fails to support rotation of western Newfoundland. Nature 266:314–318

Didenko AN, Mossakovsky AA, Pechersky DM, Ruzhentsev SV, Samygin SG, Kheraskova TN (1994) Geodynamics of Paleozoic oceans of Central Asia. Russ Geol Geophys 7–8:59–75 (in Russian)

Elming SÅ, Kravchenko SN, Layer P, Rusakov OM, Glevasskaya AM, Mikhailova NP, Bachtadse V (2007) Palaeomagnetism and $^{40}Ar/^{39}Ar$ age determinations of the Ediacaran traps from the southwestern margin of the East European Craton, Ukraine: relevance to the Rodinia break-up. J Geol Soc Lond 164:969–982

Elming S-Å, Pisarevsky SA, Layer P, Bylund G (2013) A palaeomagnetic and 40Ar/39Ar study of mafic dykes in southern Sweden: a new Early Neoproterozoic key-pole for the Baltic Shield and implications for Sveconorwegian and Grenville loops. Precambr Res. doi:10.1016/j.precamres.2013.12.007

Buchan KL, Hamilton MA, Davidson A, Ernst, RE (2011) U–Pb geochronology and magnetic overprinting of late Neoproterozoic dyke swarms associated with Ottawa Bonnechere and Timiskaming grabens and SW faults of the upper St. Lawrence river, GAC/MAC-SEG-SGA Joint Annual Meeting, 25–27 May (Ottawa 2011, Negotiating Past and Future Change), abstract vol 34, p 29

Ernst RE, Buchan KL (1997) Giant radiating dyke swarms' their use in identifying pre-mesozoic large igneous provinces and mantle plumes. AGU Geophys Mon 100:297–333

Evans DAD (2003) A fundamental Precambrian–Phanerozoic shift in earth's glacial style? Tectonophysics 375:353–385

Fisher RA (1953) Dispersion on a sphere. Proc R Soc Lond Ser A Math Phys Sci 217(1130):295–305

Grazhdankin DV, Marusin VV, Meert J, Krupenin MT, Maslov AV (2011) Kotlin regional stage in the South Urals. Doklady Earth Sci 440:1222–1226

Greiling RO, Jensen S, Smith AG (1999) Vendian–Cambrian subsidence of the passive margin of western Baltica—application of new stratigraphic data from the Scandinavian Caledonian margin. Nor Geol Tidsskr 79:133–144

Halls H, Hamilton M (2012) Magnetochronology of the Neoproterozoic Grenville dyke swarm. 34th international geological congress, Brisbane 2012, abstract 2925

Hartz EH, Torsvik TH (2002) Baltica upside down: a new plate tectonic model for Rodinia and the Iapetus Ocean. Geology 30:255–258

Higgins MD, van Breemen O (1998) The age of the Sept Iles layered mafic intrusion, Canada: implications for the Late Neoproterozoic/Cambrian History of Southeastern Canada. J Geol 106:421–431

Hodych JP, Cox RA, Košler J (2004) An equatorial Laurentia at 550 Ma confirmed by Grenvillian inherited zircons dated by

LAM ICP-MS in the Skinner Cove volcanics of western Newfoundland: implications for inertial interchange true polar wander. Precambr Res 129:93–113
Hoffman PF, Schrag DP (2002) The snowball Earth hypothesis: testing the limits of global change. Terra Nova 14:129–155
Hovland A (2012) Paleomagnetiske studier av de neoproterozoiske Fugleberg og Nyborgformasjonene, Varangerhalvøyen, Finnmark. MSc thesis, Department of Earth Science, University of Bergen, Norway (in Norwegian)
Iglesia Llanos MP, Tait JA, Popov V, Ablamasova A (2005) Palaeomagnetic data from Ediacaran (Vendian) sediments of the Arkhangelsk region, NW Russia: an alternative apparent polar wander path of Baltica for the Late Proterozoic–Early Palaeozoic. Earth Planet Sci Lett 240:732–747
Iosifidi AG, Khramov AN, Bachtadse V (2005) Multicomponent magnetization of Vendian sedimentary rocks in Podolia, Ukraine. Russ J Earth Sci 7(1):1–14
Kamo SL, Gower CF (1994) U–Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Range dykes, southeastern Labrador. Atl Geol 30:259–262
Khramov AN (ed) Palaeomagnetic directions and palaeomagnetic poles. Reference book for the USSR. Materials of the WDC-B, Moscow (issue 1–1971, issue 2–1973, issue 3–1975, issue 4–1979, issue 5–1982, issue 6–1986, issue 7–1989) (in Russian)
Kirschvink JL (1980) The least squares line and plane and the analysis of palaeomagnetic data. Geophys J R Astron Soc 62:699–718
Kirschvink JL (1992) Late Proterozoic low-latitude global glaciation: the snowball Earth. In: Schopf JW, Klein C (eds) The Proterozoic biosphere. Cambridge University Press, New York, pp 51–52
Knoll AH (1992) The early evolution of eukaryotes, a geological perspective. Science 256:622–637
Knoll AH, Walter MR, Narbonne GM, Christie-Blick N (2006) The Ediacaran Period: a new addition to the geologic time scale. Lethaia 39:13–30
Kuznetsov NB, Shazillo AV (2011) The first finds of skeletal fossils in the Kuk Karauk Formation of the Asha Group (Southern Urals) and their significance for determining the beginning of the Pre-Uralian–Timanian orogeny. Doklady Earth Sci 440:1239–1244
Levashova NM, Bazhenov ML, Meert JG, Kuzhetsov NB, Golovanova IV, Danukalov KN, Fedorova NM (2013) Paleogeography of Baltica in the Ediacaran: paleomagnetic and Geochronological data from the clastic Zigan Formation, South Urals. Precambr Res 236:16–30
Li ZX, Bogdanova SV, Collins A, Davidson A, De Waele B, Ernst RE, Fitzsimons I, Fuck R, Gladkochub D, Jacobs J, Karlstrom K, Lu S, Milesi J-P, Myers J, Natapov L, Pandit M, Pease V, Pisarevsky SA, Thrane K, Vernikovsky V (2008) Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambr Res 160:179–210
Lubnina NV (2009) Position of the East European continent during the Late Paleozoic time: analysis of remagnetization of rocks in the Southern Urals. Explor Prot Miner Resour 12:34–42 (in Russian)
Mardia KV (1972) Statistics of directional data. Academic Press, London
Martin MW, Grazhdankin DV, Bowring SA, Evans DAD, Fedonkin MA, Kirschvink JL (2000) Age of Neoproterozoic bilatarian body and trace fossils, White Sea, Russia: implications for metazoan evolution. Science 288:841–845
Maslov AV, Isherskaya MV (2002) Riphean sedimentary sequences of the eastern and northeastern margins of the Eastern European craton. Russ J Earth Sci 4:271–276
Matenaar I, Glasmacher UA, Pickel W, Giese U, Pazukhin VN, Kozlov VI, Puchkov VN, Stroink L, Walter R (1999) Incipient metamorphism between Ufa and Beloretzk, western fold-and-thrust belt, southern Urals, Russia. Geol Rundsch 87:545–560
McCall GJH (2006) The Vendian (Ediacaran) in the geological record: enigmas in geology's prelude to the Cambrian explosion. Earth Sci Rev 77:1–299
McCausland PJA, Hodych JP (1998) Palaeomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and opening of the Iapetus Ocean. Earth Planet Sci Lett 163:15–29
McCausland PJA, Murphy JB, MacNiocaill C (2006) Endings and beginnings: paleogeography of the Neoproterozoic–Cambrian transition. Precambr Res 147:187–192
McCausland PJA, Hankard F, Van der Voo R, Hall CM (2011) Ediacaran paleogeography of Laurentia: paleomagnetism and 40Ar–39Ar geochronology of the 583 Ma Baie des Moutons syenite, Quebec. Precambr Res 187:58–78
McFadden PL, McElhinny MW (1990) Classification of the reversal test in palaeomagnetism. Geophys J Int 103(3):725–729
Meert JG (2014) Ediacaran–Early Ordovician paleomagnetism of Baltica: a review. Gondwana Res 25(1):159–169
Meert JG, Van der Voo R (2001) Comment: New palaeomagnetic result from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian by S. A. Pisarevsky R. A. Komissarova A. N. Khramov. Geophys J Int 146:867–870
Meert JG, Van der Voo R, Payne TW (1994) Paleomagnetism of the Catoctin volcanic province: a new Vendian–Cambrian apparent polar wander path for North America. J Geophys Res 99:4625–4641
Meert JG, Torsvik TH, Eide EA, Dahlgren S (1998) Tectonic significance of the Fen Province, S. Norway: constraints from geochronology and paleomagnetism. J Geol 106:553–564
Meert JG, Walderhaug HJ, Torsvik TH, Hendricks BWH (2007) Age and palaeomagnetic signature of the Alnø carbonatite complex (NE Sweden): additional controversy for the Neoproterozoic position of Baltica. Precambr Res 154:159–174
Murphy JB, Pisarevsky SA, Nance RD, Keppie JD (2004) Neoproterozoic–early Paleozoic evolution of peri-Gondwanan terranes: implications for Laurentia–Gondwana connections. Int J Earth Sci 93:659–692
Murphy JB, Pisarevsky SA, Nance RD (2013) Potential geodynamic relationships between the development of peripheral orogens along the northern margin of Gondwana and the amalgamation of West Gondwana. Mineral Petrol. doi:10.1007/s00710-012-0207-9
Murthy G, Gower CF, Turbett M, Pätzold R (1992) Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador, Canada. Can J Earth Sci 29:1224–1234
Narbonne G (2008) The Gaskiers glaciation as a significant divide in Ediacaran history and stratigraphy. 33rd international geological congress, abstract 1342961, Oslo
Nawrocki J, Boguckij A, Katinas V (2004) New late Vendian palaeogeography of Baltica and TESZ. Geol Q 48:309–316
Nikishin AM, Ziegler PA, Stephenson RA, Cloetingh SAPL, Furne AV, Fokin PA, Ershov AV, Bolotov SN, Korotaev MV, Alekseev AS, Gorbachev VI, Shipilov EV, Lankreijer A, Bembinova EYu, Shalimov IV (1996) Late Precambrian to Triassic history of the East European Craton: dynamics of sedimentary basin evolution. Tectonophysics 268:23–63
Nosova AA, Veretennikov NV, Levskii LK (2005) Nature of the mantle source and specific features of crustal contamination of neoproterozoic flood basalts of the Volhynia Province (Nd–Sr isotopic and ICP-MS geochemical data). Doklady Earth Sci 401(3):429–433
Pimentel MM, Fuck RA, Botelho NF (1999) Granites and the geodynamic history of the Neoproterozoic Brasilia belt, Central Brazil: a review. Lithos 46:463–483
Pisarevsky SA (2005) New edition of the global paleomagnetic database. EOS Trans 86(17):170

Pisarevsky SA, Komissarova RA, Khramov AN (2000) New palaeomagnetic results from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian. Geophys J Int 140:598–610

Pisarevsky SA, Komissarova RA, Khramov AN (2001) Reply to the comment of J.G.Meert and R.Van der Voo "new palaeomagnetic results from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian". Geophys J Int 146:871–873

Pisarevsky SA, Wingate MTD, Powell CMcA, Johnson S, Evans DAD (2003) Models of Rodinia assembly and fragmentation. In: Yoshida M, Windley B, Dasgupt S (eds) Proterozoic East Gondwana: supercontinent assembly and breakup. Geological Society of London Special Publication 206, pp 35–55

Pisarevsky SA, Murphy JB, Cawood PA, Collins AS (2008) Late Neoproterozoic and Early Cambrian palaeogeography: models and problems. In: Pankhurst RJ, Trouw RAJ, Brito Neves BB, de Wit MJ (eds) West Gondwana: pre-Cenozoic correlations across the South Atlantic region. Geological Society of London Special Publications 294, pp 9–31

Popov V, Iosifidi A, Khramov A, Tait J, Bachtadse V (2002) Paleomagnetism of upper Vendian sediments from the Winter Coast, White Sea region, Russia: implications for the paleogeography of Baltica during Neoproterozoic times. J Geophys Res 107(B11, 2315). doi:10.1029/2001JB001607

Popov V, Khramov A, Bachtadse V (2005) Palaeomagnetism, magnetic stratigraphy, and petromagnetism of the Upper Vendian sedimentary rocks in the sections of the Zolotitsa River and in the Verkhotina Hole, Winter Coast of the White Sea, Russia. Russ J Earth Sci 7(2):1–29

Puchkov VN (2003) Magmatism on passive margins of the Rodinia. Materials of the international conference. Ufa, Geological Institute of the USC RAS, 40 pp

Puchkov VN (2010) Geology of the Urals and the Priuralie (Actual Problems of Stratigraphy, Tectonics, Geodynamics and Metallogeny). Ufa: Disain Poligraf Servis. 281 p (in Russian)

Puchkov VN (2012a) On the age of the Asha series, South Urals. In: Puchkov VN (ed) Geology, mineral deposits and problems of paleoecology of Bashkortostan, Urals and surrounding areas. Materials of the 9th inter-regional scientific and practical conference. Ufa, 19–22 November 2012, pp 47–51 (in Russian)

Puchkov VN (2012b) Dike swarms and related igneous complexes in the Urals. Geotectonics 46:37–46

Puchkov VN, Bogdanova SV, Ernst R, Kozlov VI, Krasnobaev AA, Söderlund U, Wingate MTD, Postnikov AV, Sergeeva ND (2013) The ca. 1380 Ma Mashak igneous event of the Southern Urals. Lithos 174:109–124

Puffer JH (2002) A Late Neoproterozoic eastern Laurentian superplume: location, size, chemical composition, and environmental impact. Am J Sci 302:1–27

Roberts D, Siedleska A (2002) Timanian orogenic deformation along the northeastern margin of Baltica, Northwest Russia and Northeast Norway, and Avalonian–Cadomian connections. Tectonophysics 352:169–184

Shipunov SV, Chumakov NM (1991) Paleomagnetism of Upper Proterozoic deposits of the Kola Peninsula. Geotectonics 25:401–410

Shipunov SV, Shatsillo AV, Orlov SYu (2007) Validity of paleomagnetic poles and principles of constructing their wander paths: a case study of the East European platform. Izvestiya Phys Solid Earth 43(11):960–966

Siedlecka A, Roberts D, Nystuen JP, Olovyanishnikov VG (2004) Northeastern and northwestern margins of Baltica in Neoproterozoic time: evidence from the Timanian and Caledonian Orogens. In: Gee DG, Pease VL (eds) The Neoproterozoic Timanide Orogen of Eastern Baltica. Geological Society London Memoirs 30, pp 169–190

Smethurst MA, Khramov AN, Pisarevsky SA (1998) Palaeomagnetism of the Lower Ordovician Orthoceras Limestone: implications for drift history for Baltica in the early Palaeozoic and for magnetostratigraphy. Geophys J Int 133:44–56

St Seymour K, Kumarapeli PS (1995) Geochemistry of the Grenville Dyke Swarm: role of plume-source mantle in magma genesis. Contrib Mineral Petrol 120:29–41

Stukas V, Reynolds PH (1974) 40Ar/39Ar dating of the Long Range dikes, Newfoundland. Earth Planet Sci Lett 22:256–266

Symons DTA, Chiasson AD (1991) Paleomagnetism of the Callander Complex and the Cambrian apparent polar wander path for North America. Can J Earth Sci 28:355–363

Tanczyk EI, Lapointe P, Morris WA, Schmidt PW (1987) A paleomagnetic study of the layered mafic intrusion at Sept-Iles, Quebec. Can J Earth Sci 24:1431–1438

Tohver E, D'Agrella-Filho MS, Trindade RIF (2006) Paleomagnetic record of Africa and South America for the 1200–500. Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambr Res 147:193–222

Van Alstine DR, Gillett SL (1979) Paleomagnetism of Upper Precambrian sedimentary rocks from the Desert Range, Nevada. J Geophys Res 84:4490–4500

Van der Voo R (1990) The reliability of paleomagnetic data. Tectonophysics 184:1–9

Walderhaug HJ, Torsvik TH, Halvorsen E (2007) The Egersund dykes (SW Norway): a robust Early Ediacaran (Vendian) palaeomagnetic pole from Baltica. Geophys J Int 168:935–948

Willner AP, Ermolaeva T, Stroink L, Glasmacher UA, Giese U, Puchkov VN, Kozlov VI, Walter R (2001) Contrasting provenance signals in Riphean and Vendian sandstones in the SW Urals (Russia): constraints for a change from passive to active continental margin conditions in the Neoproterozoic. Precambr Res 110:215–239

Willner AP, Sindern S, Metzger R, Ermolaeva T, Kramm U, Puchkov V, Kronz A (2003) Typology and single grain U/Pb ages of detrital zircons from Proterozoic sandstones in the SW Urals (Russia): early time marks at the eastern margin of Baltica. Precambr Res 124:1–20